

**Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Laboratory Exercise 3
Schematic Design Library**

Due: 02/07/2025

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1 Introduction

The goal of this lab is for students to get started on building their own cell library. This involves creating separate schematic cells for each of the following gates: 2 input NAND, 2 input NOR, 2-1 MUX, 2 input XOR in addition to testing the 3 input AOI schematic created in lab exercise 1.

There is **no lab report for this lab**.

Follow instructions from section 4 for deliverables.

Keep the following in mind as you work through this lab.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES ARE LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

`/classes/ee620/bin/unlock_cadence.csh`

- **Hotkeys:** You should learn how to use the hotkeys for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

Options → Bindkey

Select a category (e.g. **schematic**) and a list of current hotkeys (bindkeys) will be shown for it.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!
- DO NOT delete any Cadence files from command line or file manager!

2 Procedure

When starting this lab, make sure you are on a fresh logged-in session. If not, then close Cadence, log out and log back in. Open the Cadence tools. Now create separate cells for a NAND, NOR, 2-1 MUX, and XOR. Name them using the same naming convention as your inverter (i.e., using your initials "FML_") with each of the following: NAND2X1, NOR2X1, MUX2X1, XOR2X1 (i.e., FML_NAND2X1). Draw the schematics shown in the Figures (2, 3, 4, and 5) and follow the same procedure you learned in previous labs to create the schematics. **Also do not forget to create symbols for all of them.** Creating a Verilog model for each schematic is optional, but encouraged.

Before you get started...

- Remember to size transistors for equal pull-up and pull-down drive strength. Keep in mind that the minimum width for the 45 nm technology is 120 nm. The unit inverter aspect ratios specific to this technology are:

$$\frac{W}{L}_{NMOS} = 120/45 \quad (1)$$

$$\frac{W}{L}_{PMOS} = 240/45 \quad (2)$$

- When setting input pins you don't need to put multiple pins! Instead, **create the necessary input pins at the top corner then connect a little loose wire**. Now, press **L** on your keyboard to create labels. You will label these the same as the pin names. Then add little wires on each transistor and label them to their corresponding matching name. A small example is shown in Figure 1.

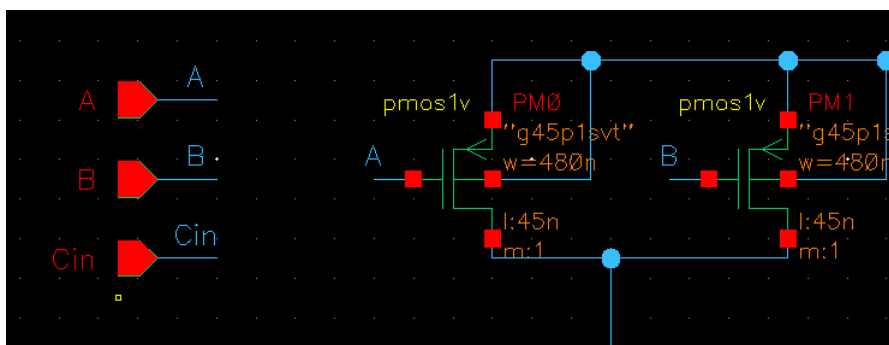


Figure 1: Adding Pins With Labels

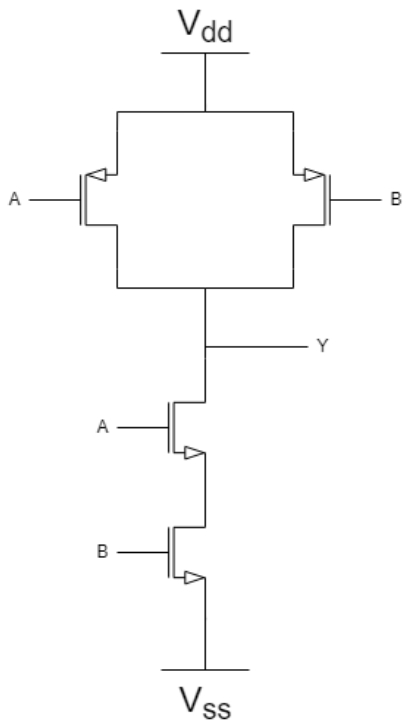


Figure 2: 2 Input NAND Schematic

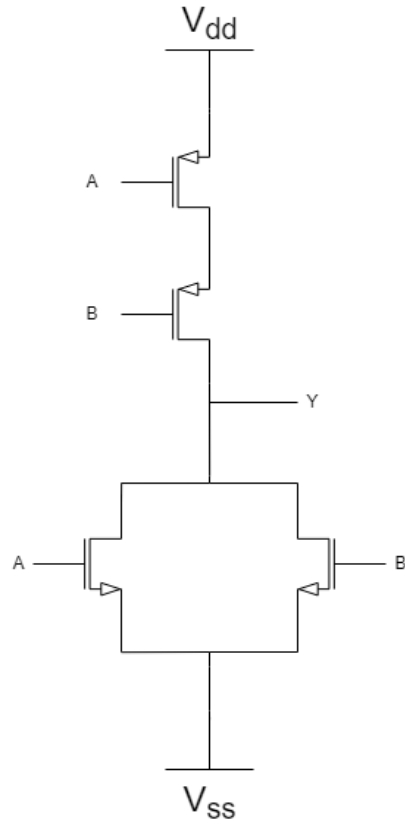


Figure 3: 2 Input NOR Schematic

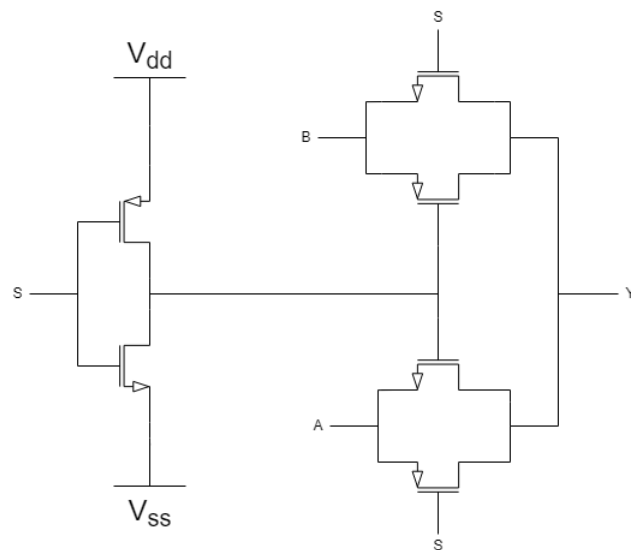


Figure 4: 2-1 MUX Schematic

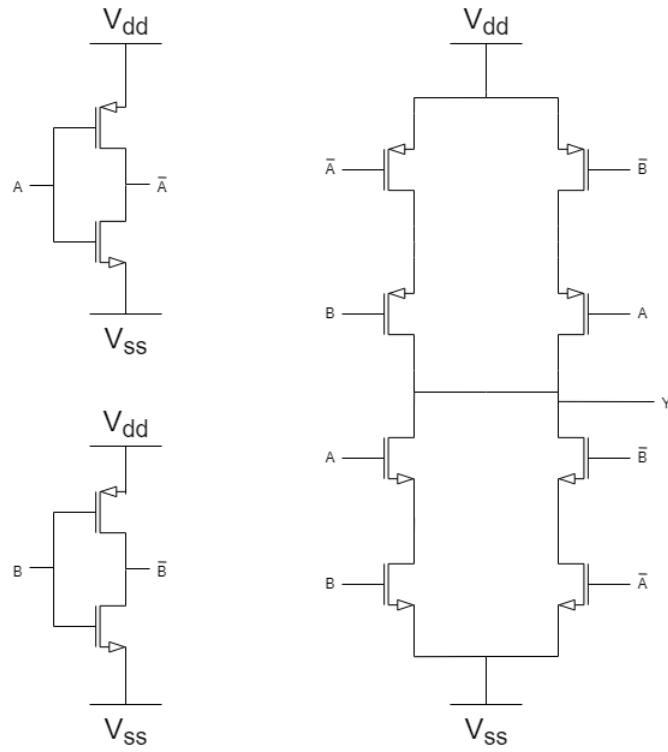


Figure 5: 2 Input XOR Schematic

3 Testing

To demonstrate functionality for all schematic cells, you will create a testbench for each design. You will then use ADE XL to perform transient simulation and measure timing values (rise time, fall time, T_{PLH}, T_{PHL}) for each design, similar to what we have done in lab exercise 2. **Note** that the stimuli and calculations for each cell testbench will be different from those you configured for exercise 2, as the internal logic for each gate is different. For the AOI cell, use the schematic created during lab exercise 1. Here is an example of what testing will consist of:

For the NAND2

- $A \rightarrow Y \uparrow$ (you may name it: A_Y_Rise)
 - RiseTime function
 - Rising (Low to High) Propagation Delay function
- $A \rightarrow Y \downarrow$ (you may name it: A_Y_Fall)
 - FallTime function
 - Falling (High to Low) Propagation Delay function
- $B \rightarrow Y \uparrow$ (you may name it: B_Y_Rise)
 - RiseTime function
 - Rising (Low to High) Propagation Delay function
- $B \rightarrow Y \downarrow$ (you may name it: B_Y_Fall)
 - FallTime function
 - Falling (High to Low) Propagation Delay function
- Functional/Transient Simulation (This is the same set up as lab exercise 2)

For each cell, we will test every possible case (tracing each path separately) using the truth table. Based on the truth table we can decide what single pin we have to toggle to cause the output to change. **This means, when you test a certain path (i.e $A \rightarrow Y \uparrow$) you will make B path static 1 and vice versa.** Below is an example of the necessary configurations (note that V1 and V2 will change depending on your path):

- $A \rightarrow Y \uparrow$ (this is the $A = 1$ and $B = 1$ case in the truth table)

Go to **Test Editor → Stimuli → Inputs**. Make sure to **Enable** the inputs.

Now fill only the following for A:

Function: Pulse
Voltage 1: 1
Voltage 2: 0
Delay Time: 100ns

Rise time: 100ps

Fall time: 100ps

Now fill only the following for B:

Function: DC

DC Voltage: 1

Then create expressions for RiseTime and Rising Propagation Delay (T_{PLH})

- T_{PLH} measures time from falling edge of input A to the rising edge of output Y

These are the parameters you will be using for all tests, we will be testing only one case similar to the worst case in exercise 2.

- CAP : 100f
- Temperature : 125c
- Input Voltage: 1V
- Test Editor Global Sources is the same as usual.

Tips and tricks to make your life easier:

- Remember that you can copy tests within the same cell test environment! For each gate you may create one test case and then copy it. With this, all you need to do is change the stimuli and view your “output setup” to modify the calculator syntax to whatever operation you are trying to do. This reduces your chances of using the calculator over and over again.
- You can also add expressions in the “Outputs Setup” if you already know the calculator syntax (you can copy paste it). Just right click the white empty spot at the bottom of your test list and click “Add Expression”.
- You can select each test to run individually, it will take less time than having them all run every time.

The next page shows how your tests list in **Data View** should look. Note that in each test you will have two functions (i.e. RiseTime and then Propagation Delay).

1. For the NAND2

- $A \rightarrow Y \uparrow$
- $A \rightarrow Y \downarrow$
- $B \rightarrow Y \uparrow$
- $B \rightarrow Y \downarrow$
- Functional

2. For the NOR2

- $A \rightarrow Y \uparrow$
- $A \rightarrow Y \downarrow$
- $B \rightarrow Y \uparrow$
- $B \rightarrow Y \downarrow$
- Functional

3. For the MUX2

- $A \rightarrow Y \uparrow$
- $A \rightarrow Y \downarrow$
- $B \rightarrow Y \uparrow$
- $B \rightarrow Y \downarrow$
- $S \rightarrow Y \uparrow$
- $S \rightarrow Y \downarrow$
- Functional

4. For the XOR2

- $A \rightarrow Y \uparrow$
- $A \rightarrow Y \downarrow$
- $B \rightarrow Y \uparrow$
- $B \rightarrow Y \downarrow$
- Functional

5. For the AOI3

- $A \rightarrow Y \uparrow$
- $A \rightarrow Y \downarrow$
- $B \rightarrow Y \uparrow$
- $B \rightarrow Y \downarrow$
- $C \rightarrow Y \uparrow$
- $C \rightarrow Y \downarrow$
- Functional

4 Datasheet

Complete the datasheets for all designs. The template is provided on MyCourses. You should have a total of 5 data sheets, one per logic gate (see Figure 6 for an example). Anything that says **layout** you may leave blank for now (these spots will be filled in a future lab). Turn in all the data sheets by the lab due date listed in MyCourses. You do not need to have the 'Datasheet' and 'Question' sign-off before the lab period, it will be graded after it is turned in. **Submit 5 datasheets (one per gate) and the signoff sheet to MyCourses (all of them need to be submitted as .pdf or .docx)**

Cell Name:		NOR2X1
Function/Truth Table: $Y = \text{NOT} (A \text{ OR } B)$		
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0
Propagation Delay (path based):		
A –T_PLH		
A –T_PHL		
B –T_PLH		
B –T_PHL		
Output Rise Time (path based):		
A – Rise Time		
B – Rise Time		
Output Fall Time (path based):		
A – Fall Time		
B – Fall Time		

Figure 6: Cell Datasheet Example (don't forget about symbol, schematic and functional simulation waveform)

5 Sign-off Sheet

Table 1: Schematic Design Library

Students Name:

Section:

Demo	Point Value	Points Earned	Date
NAND Schematic	5		
NOR Schematic	5		
MUX Schematic	10		
XOR Schematic	10		
NAND ADE XL Tests	5		
NOR ADE XL Tests	5		
MUX ADE XL Tests	10		
XOR ADE XL Tests	10		
AOI ADE XL Tests	5		
Datasheets	35		
Total Points	100		